

Group's Name : False 9

Group Members: NASAAIE BIN NORISKAMAR (LEADER) A25CS 0118
: MUHAMMAD FAIRUZ BIN HERMAN A25CS 0267
: MUHAMMAD NAJMI SHAHMI BIN MOHD LATPI A25CS 0279
: AKMAL RAFIQUE BIN AHMAD RAPHAIE A25CS 0181

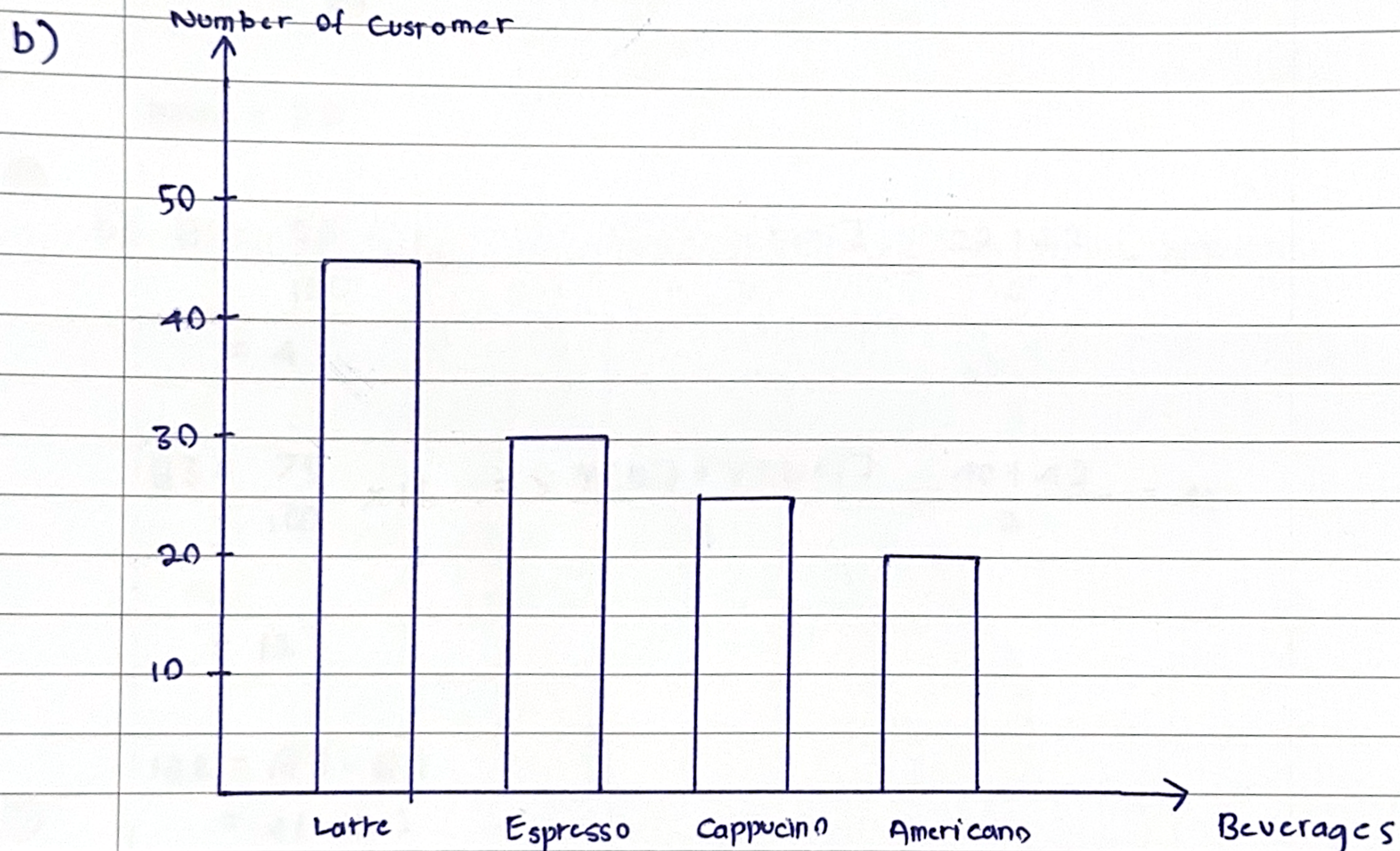
Question 1:

- a) Population = All students university who visited the library during first week
Sample = 60 students
- b) Qualitative = Student ID, Major of Study (Engineering, Arts, Business), Academic Year
Quantitative = total number of books borrowed this semester, time spent studying in the library per week, temperature of the reading room when they visit (Celsius)
- c) Discrete = total number of books borrowed this semester
Continuous = time spent studying in the library per week, temperature of the reading room (Celsius)
- d) Nominal = ~~Student ID~~ Major of Study
Ordinal = Academic Year
Interval = temperature of the reading room (Celsius)
Ratio = ~~time spent studying in the library per week~~, total number of books borrowed this semester

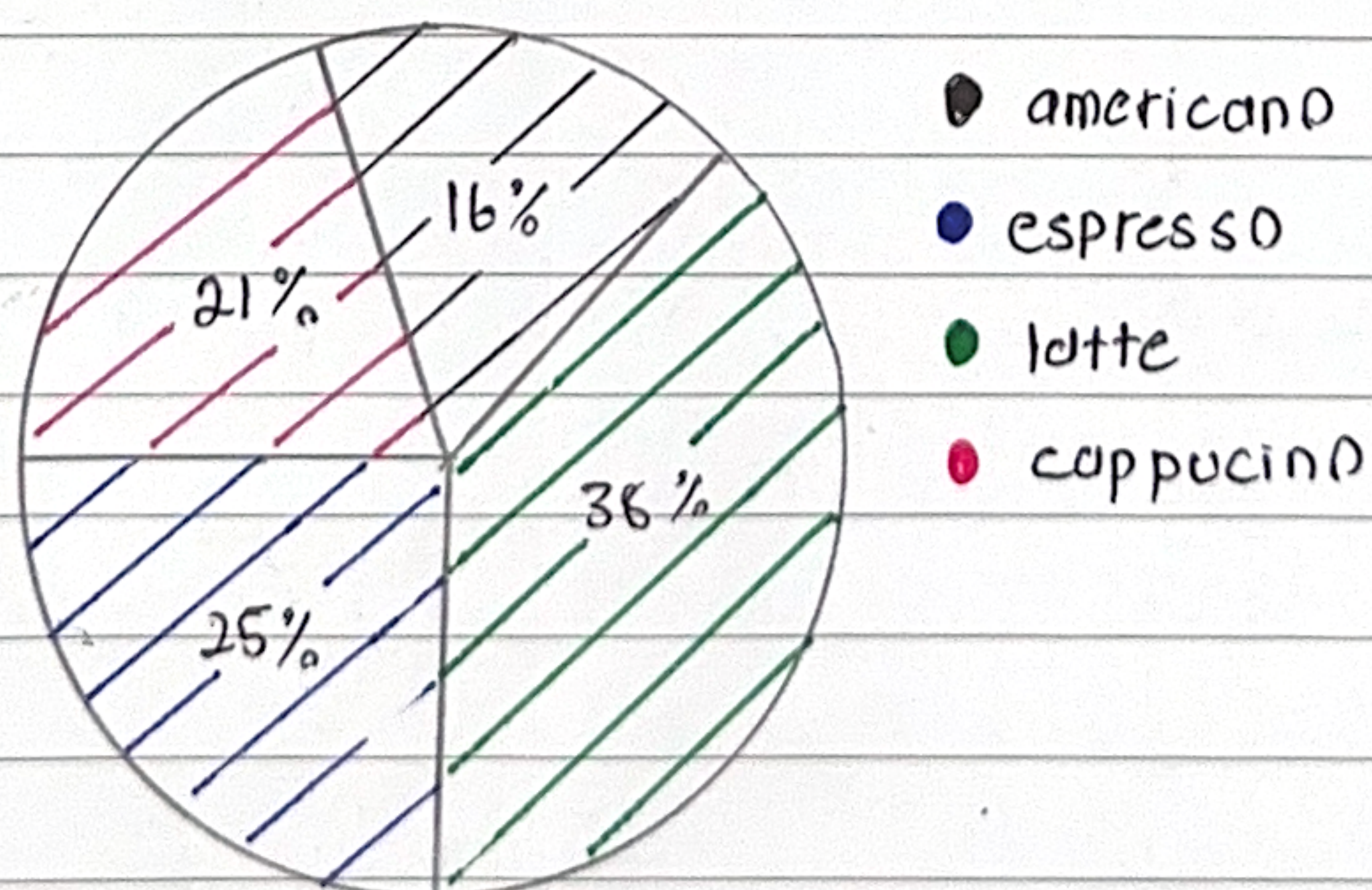
Question 2

a)

Beverages	Frequency	Relative Frequency	Cumulative Relative Frequency (CF)
Latte	45	0.38	0.38
Espresso	30	0.25	0.63
Cappuccino	25	0.21	0.84
Americano	20	0.16	1.00



c)



Question 3

$$a) \text{ min median} = \frac{15+18+20+22+22+25+26+28+30+32+35+40+42+65+70+75}{16} \\ = 35.31$$

$$\text{median} = \frac{28+30}{2} \\ = 29$$

$$\text{mode} = 22$$

$$b) Q_1 = \frac{25}{100} \times 16 \Rightarrow \frac{Y[4] + Y[4+1]}{2} = \frac{22+22}{2} = 22 \\ = 4$$

$$Q_3 = \frac{75}{100} \times 16 \Rightarrow \frac{Y[12] + Y[12+1]}{2} = \frac{40+42}{2} = 41 \\ = 12$$

$$IQR = Q_3 - Q_1 \\ = 41 - 22 \\ = 19$$

$$c) \text{ lower limit} = Q1 - 1.5(IQR)$$

$$= 22 - 1.5(19)$$

$$= -6.5$$

$$\text{upper limit} = Q3 - 1.5(IQR)$$

$$= 41 - 1.5(19)$$

$$= 69.5$$

$$70, 75 > 69.5$$

$$\text{Outliers} = 70, 75$$

$$3 \times IQR = 3 \times 19 = 57$$

$$Q3 + 3(IQR) = 41 + 57 = 98$$

$$\left. \begin{array}{l} 70 < 98 \\ 75 < 98 \end{array} \right\} \text{mild outliers}$$

d)

